UAI201: Programming Essentials for AI and ML

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Introduction to Python programming: Setting up the Python environment (IDEs, Jupyter Notebooks), Writing and running Python scripts, Basic syntax, variables, and data types (integers, floats, strings)
Control Structures: `if`, `else`, `elif`, Loops: `for`, `while`, `break`, `continue`, `pass`
Data Structures in Python: Lists and list operations (indexing, slicing, appending), Tuples and their immutability, Dictionaries: key-value pairs, adding and removing elements, Sets: basic operations and set methods
Working with Strings and Files: String manipulation and formatting, Reading from and writing to files, Working with file paths, Handling exceptions in file operations. Using numpy, torch, mat and pickle files
Error Handling and Debugging: Understanding exceptions and error types, Using `try`, `except`, `else`, `finally` for error handling, Debugging techniques and tools.
Libraries and APIs: Introduction to Python libraries (e.g., `requests`, `json`, `datetime`), Working with APIs to fetch and process data, Understanding JSON data format, Manipulating XML files. Introduction to Data Handling with Pandas: Introduction to the Pandas library, Working with Data Frames: loading, selecting, and filtering data, Basic data analysis and manipulation.

Text Books / Reference Books:

1. A. Sweigart, *Automate the Boring Stuff with Python: Practical Programming for Total Beginners*, 1st ed. No Starch Press, 2015.
2. E. Matthes, *Python Crash Course*, 2nd ed. No Starch Press, 2019.
3. M. Lutz, *Learning Python*, 5th ed. O'Reilly Media, 2013.

Course Learning Objectives

1. Understand and Apply Basic Programming Concepts
2. Implement Control Structures and Data Handling Techniques
3. Perform String Manipulation, File Operations, and Error Handling
4. Utilize Python Libraries for Data Handling and API Interaction

SEMESTER III